TriNova

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Domain and Problem Statement

❖Domain

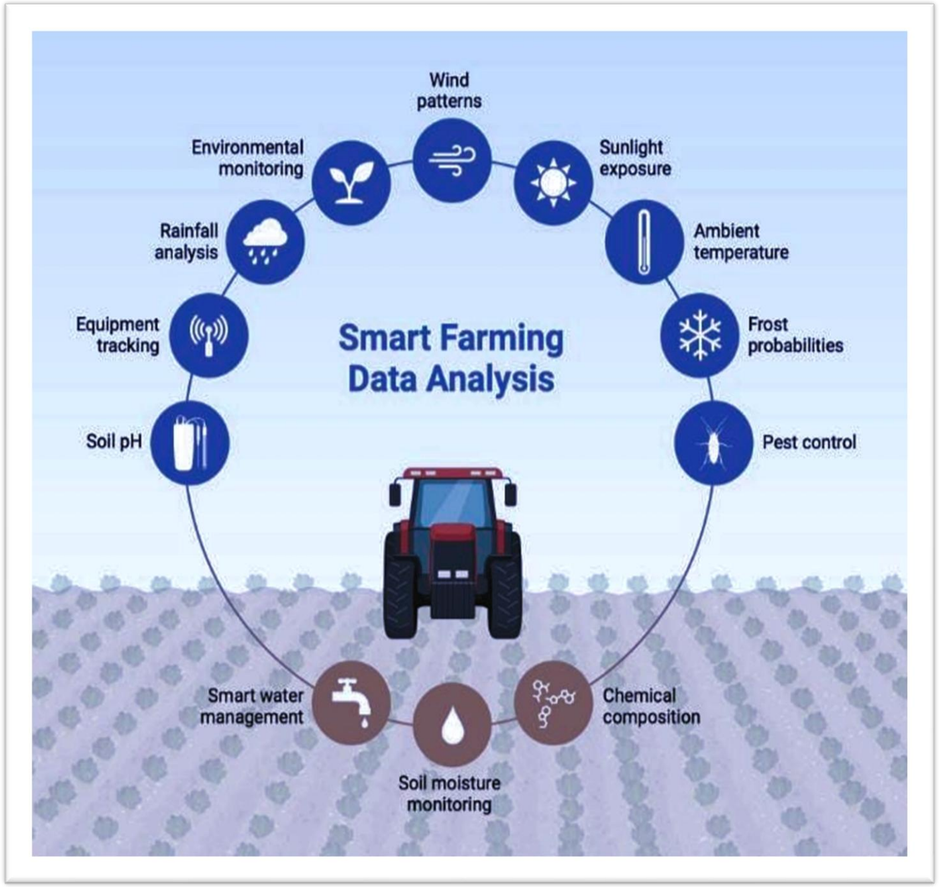
Agriculture | Artificial Intelligence | Resource Optimization

❖Problem Statement

Farmers often choose crops without accurately considering soil quality, weather conditions, water availability, fertilizer requirements, and budget constraints. This results in lower productivity and unnecessary resource wastage.

❖Our Solution

AgriSmart AI recommends suitable crops, predicts expected yield, and optimizes land, water, fertilizer, and budget allocation to maximize profit while minimizing waste.



Idea/Solution

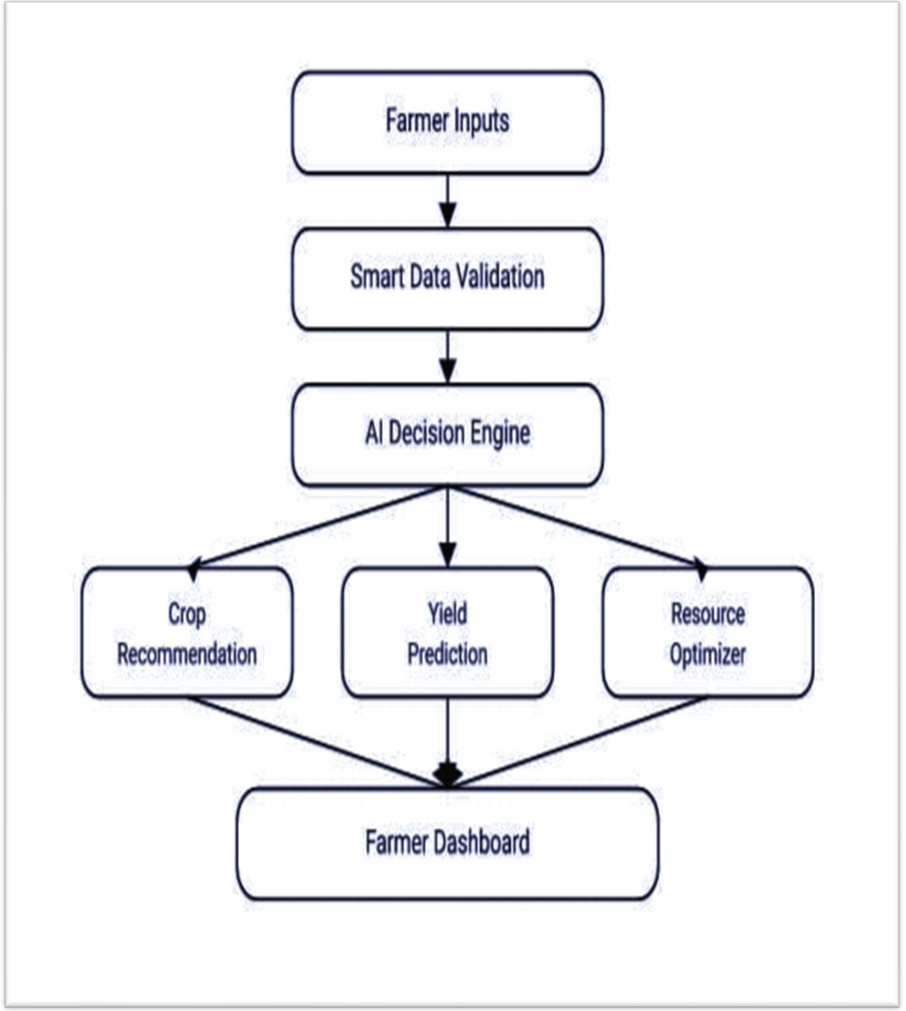
❖ AgriSmart AI

An intelligent farming assistant that provides data-driven crop planning.

❖ How It Works

1.Farmer enters soil, weather, and budget details.
2.AI validates the data.
3.Crop Recommendation Model selects the best 3 crops.
4.Yield Prediction estimates production.
5.Resource Optimizer allocates land, water, and fertilizer.
6.Dashboard displays profit, yield, and resource savings.

WORK FLOW



Existing Vs Proposed System

Existing System	AgriSmart AI
Manual crop selection	AI-based recommendations
Experience-based decisions	Data-driven decisions
Fixed resource planning	Smart optimization
No yield prediction	Expected yield prediction
No profit estimation	Profit calculator
Resource wastage	Optimized allocation

Innovation & USP

❖ Innovation

1. AI-Based Crop Matching
2. Smart Land & Water Optimization
3. Budget & Fertilizer Optimization
4. What-If Farming Simulation
5. Profit-Oriented Crop Planning
6. Explainable AI Recommendations
7. Telugu + English Farmer Support



❖ What makes AgriSmart AI unique?

1.AI-powered crop recommendation.
2.Smart allocation of land, water, fertilizer, and budget.
3.Profit-focused farming plans.
4.What-if simulation (less water, different budget).
5.Telugu + English support.
6.Farmer-friendly dashboard.
7.Offline demo capability.

Objectives & Scope of solution

❖ Objectives

1.Recommend suitable crops.
2.Predict expected yield.
3.Optimize agricultural resources.
4.Reduce water and fertilizer wastage.
5.Increase farmer profitability.

❖ Scope

1.Small and medium farmers.
2.Seasonal crop planning.
3.Multi-crop comparison.Scalable for different regions.



System Architecture

❖ Architecture Flow

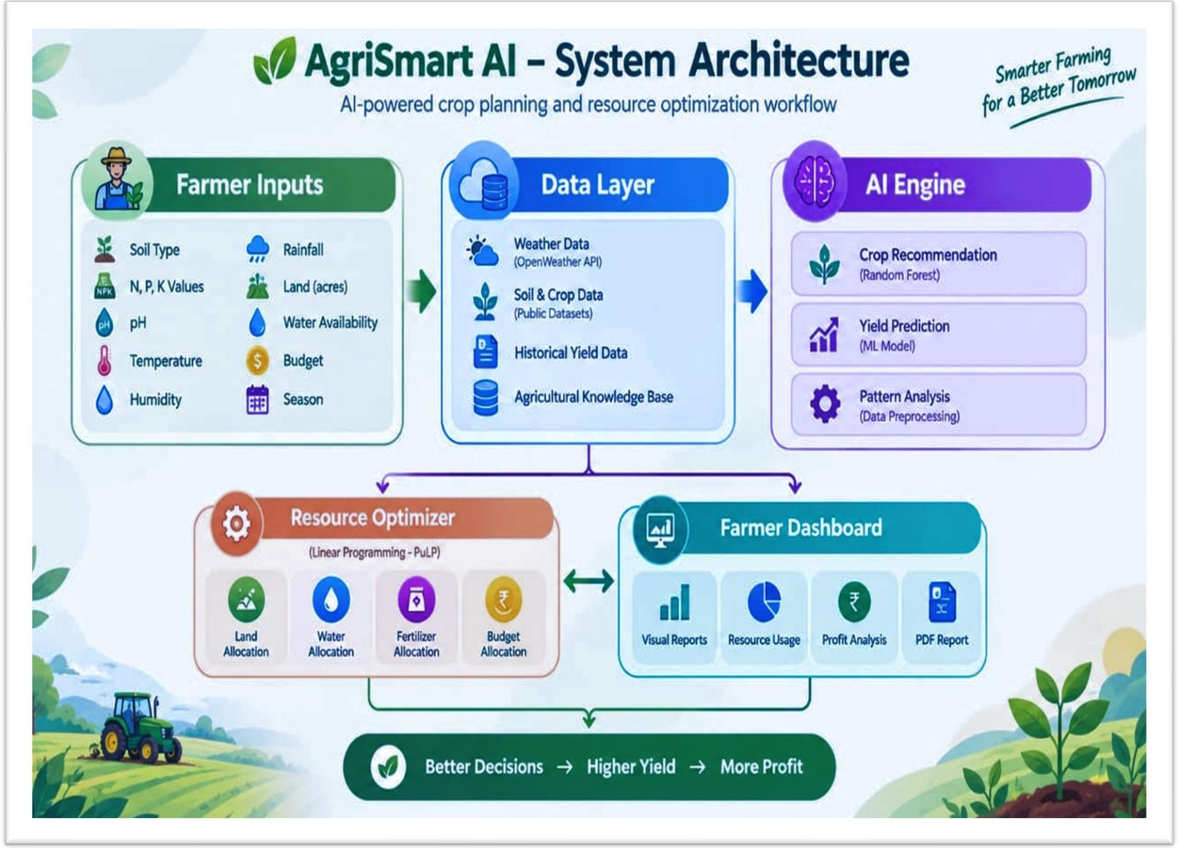
Farmer Inputs → Data Layer → AI Engine → Resource Optimizer → Dashboard → Farmer Decision Support

❖ AI Engine

1.Random Forest (Crop Recommendation).
2.Yield Prediction Model3.Data Preprocessing.

❖ Output

1.Best Crop
2.Expected Yield
3.Resource Allocation
4.Profit Report



Resources & Required

❖ Technologies

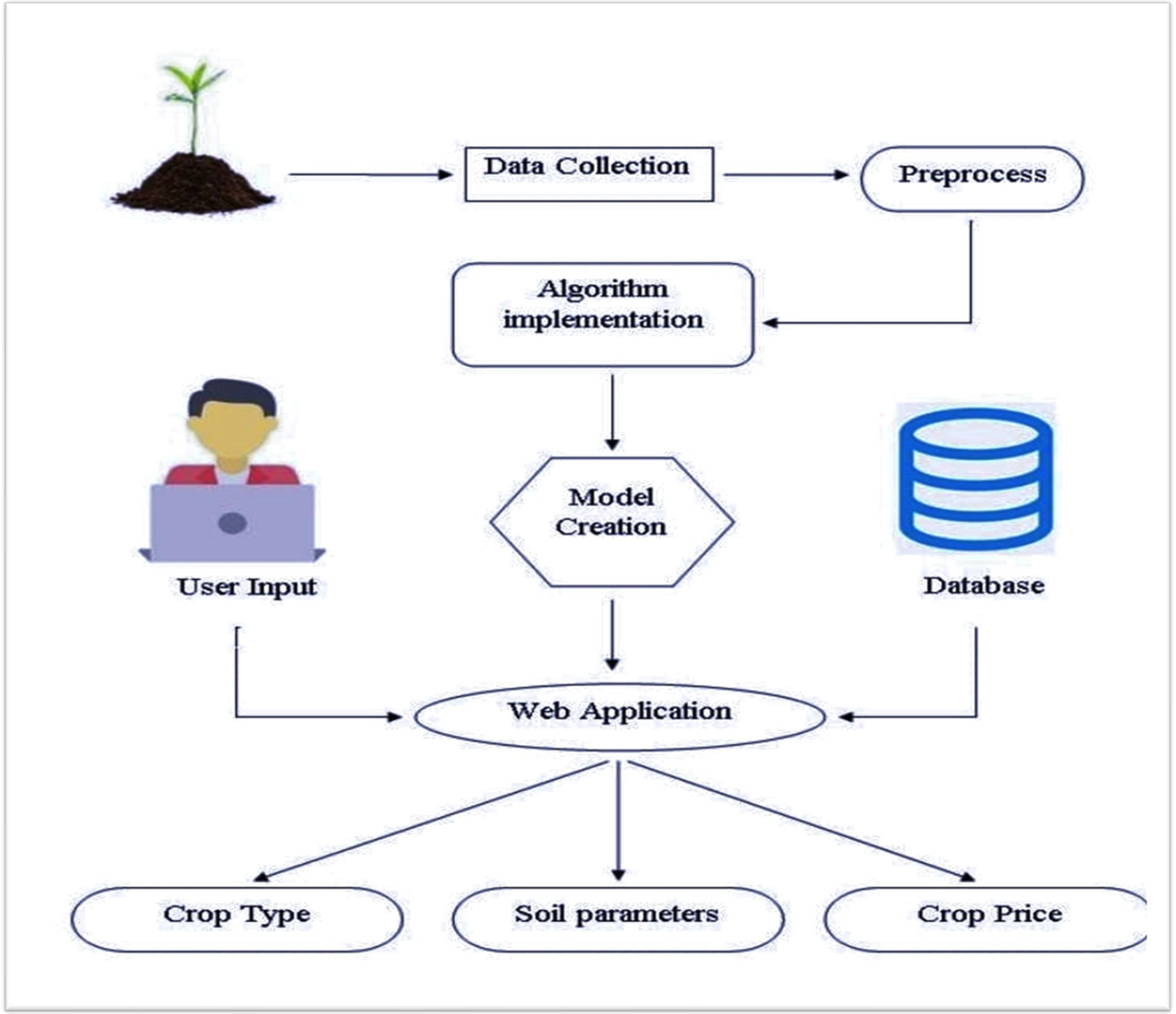
1. Python.
2. Flask.
3. HTML, CSS, JavaScript.
4. Scikit-learn.
5. PuLP.
6. SQLite / MySQL.

❖ Datasets

1. Crop Recommendation Dataset.
2. Soil NPK Data.
3. Historical Yield Data.
4. Weather Data.

❖ Tools

1. Google AI Studio
2. VS Code
3. GitHub
4. OpenWeather API



Thank You!

AgriSmart AI

Smart Crops | Smart Resources | Smarter Farming.

